

ASTANA IT UNIVERSITY
PROBATION PERIOD 1 REPORT

Development of a Mobile Application for Potential Allergen Detection Based on Hybrid Food Ingredient Analysis

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1. Project Proposal

1.1 Project Information

Project Overview:
TazaScan is an existing mobile food analysis application that allows users to identify packaged food products and view information about their ingredients, nutritional values, TazaScore, Nutri-Score, and Halal Check. The diploma project extends the existing system with a new module for detecting potential food allergens through hybrid ingredient analysis.

1.2 Problem Statement

Food product labels usually provide an ingredient list, but this information is not always easy to interpret from the point of view of potential allergens. An allergenic component may appear under a specific ingredient name or as part of a more complex formulation, so simply displaying the ingredient list does not directly help the user understand which components may require attention.

The current version of TazaScan already allows users to identify products by barcode, text search, or package photo and view nutrition information, ingredients, TazaScore, Nutri-Score, and Halal Check. However, allergen information is not analyzed automatically. At the moment, the application only displays a general notice when allergen information is unavailable.

Therefore, the diploma project focuses on adding a separate allergen analysis capability to the existing TazaScan system. The main problem is how to analyze the available ingredient information and identify potential allergens while also showing the user which ingredient caused the result. The project does not focus on personalized nutrition recommendations or meal planning.

1.3 Target Users

The primary users of TazaScan are consumers who want to quickly check the composition and nutritional information of packaged food products. The allergen analysis functionality is intended especially for users who need to pay attention to potentially allergenic ingredients and may find ingredient labels difficult to interpret.

Users can access product information by scanning a barcode, searching by product name, or using a photo of the package. A secondary user group consists of administrators who manage product information and review product submissions through the web administration panel.

1.4 Aim

The aim of the project is to develop and integrate into TazaScan a module for detecting potential food allergens through hybrid analysis of product ingredient information and to provide users with an understandable explanation of the detected results.

1.5 Objectives

1. To analyze the current TazaScan system and define the requirements for integrating potential allergen detection into the existing product analysis workflow.
2. To define the allergen categories, ingredient representations, and preprocessing rules required for analyzing product ingredient information.
3. To design and implement a hybrid allergen detection method that combines rule-based analysis with NLP-based processing for more complex or ambiguous ingredient descriptions.
4. To integrate the allergen detection module into the existing TazaScan backend and mobile application and provide users with the detected allergen together with the ingredient evidence that triggered the result.
5. To prepare a labeled evaluation dataset containing product ingredient lists with known allergen information and use it to test the developed method.
6. To evaluate the performance of the allergen detection module using Precision, Recall, and F1-score and compare the hybrid approach with a rule-based baseline.

1.6 Expected Outcomes

By the end of the project, the following outcomes are expected:

1. A hybrid allergen detection module integrated into the existing TazaScan backend and mobile application.
2. Automatic identification of potential allergens from product ingredient information, with the ingredient evidence that caused each detection shown to the user.
3. A labeled evaluation dataset containing product ingredient lists and their known allergen information.
4. Quantitative evaluation of the developed method using Precision, Recall, and F1-score, including comparison with a rule-based baseline.
5. An updated TazaScan prototype in which allergen analysis works as part of the existing product analysis flow together with product information, ingredients, nutrition data, TazaScore, Nutri-Score, and Halal Check.
6. Technical documentation describing the allergen analysis approach, system integration, evaluation procedure, and identified limitations.

1.7 Project Scope

The project covers the development and integration of a potential allergen detection module into the existing TazaScan system.

The scope includes:

- analysis of product ingredient information already stored or obtained by TazaScan;
- preprocessing and normalization of ingredient text before allergen analysis;
- rule-based detection of clearly identifiable allergenic ingredients;
- NLP-based processing for more complex or ambiguous ingredient descriptions;
- combination of rule-based and NLP results within a hybrid detection approach;
- integration of the allergen analysis module with the existing backend and mobile application;
- presentation of detected potential allergens together with the ingredient evidence that caused the result;
- preparation of a labeled evaluation dataset;
- evaluation of the developed method using Precision, Recall, and F1-score;
- comparison of the hybrid method with a rule-based baseline.

The allergen analysis functionality will be added to the current TazaScan product workflow rather than developed as a separate application.

1.8 Out of Scope

The following functionality is outside the scope of the diploma project:

- personalized nutrition recommendations;
- automatic meal planning or diet generation;
- medical diagnosis or medical advice related to allergies;
- user health profiles or recommendations based on personal medical information;
- development of a new food scoring system to replace the current TazaScore or Nutri-Score functionality;
- full production deployment and commercial release of TazaScan;
- replacement of the existing TazaScan platform with a new application.

TazaScan will remain a broader food information platform, while the diploma project will focus specifically on potential allergen detection and its evaluation.

2. Evidence of Relevance

2.1 Method

To verify whether the identified problem is relevant to potential users, five short semi-structured discussions were conducted with people who regularly purchase packaged food products.

The discussions focused on how participants read ingredient labels, whether they experience difficulties interpreting unfamiliar ingredient names, whether they pay attention to allergens or specific ingredients they want to avoid, and whether automatic allergen identification and ingredient-level explanations would be useful.

The same seven core questions were used for all participants to make their responses comparable. Additional clarification was obtained where necessary. To preserve anonymity, the participants are identified as P1–P5.

The discussions were exploratory and intended to identify user needs rather than produce statistically representative results.

2.2 Interview Questions

1. Do you usually check the ingredient list before buying a packaged food product?
2. Do you find long or unfamiliar ingredient names difficult to understand?
3. Do you pay attention to allergens or specific ingredients that you want to avoid?
4. Have you ever been unsure whether a product contained an ingredient you wanted to avoid?
5. Would automatic identification of potential allergens be useful to you?
6. Would you prefer the application to show which ingredient caused the allergen warning?
7. What other product information do you usually consider important?

2.3 Interview Results

Participant	**Ingredient Checking**	**Difficulty Understanding Ingredients**	**Ingredients of Concern**	**Uncertainty About Ingredients**	**Allergen Detection Useful**	**Ingredient Evidence**
P1	Sometimes, especially for new products	Yes; unfamiliar names are difficult to interpret	Certain ingredients to avoid	Yes; confusing terminology	Yes	Yes; wants to know the exact ingredient
P2	Rarely, unless something	Yes; complex ingredient names	Mainly sugar	Yes; unclear packaging	Yes	Yes; wants to assess the

	seems unusual					specific ingredient
P3	Occasionally; also checks expiration dates	Yes; especially E-number additives	Carmine, pork-derived components, and additives	Yes; frequently checks for unwanted ingredients	Yes; also useful for family members	Yes
P4	Regularly, especially for unfamiliar products	Yes; complex terminology and small text	Mainly sugar	Yes; unfamiliar products and unclear labels	Yes	Yes; wants the exact ingredient responsible
P5	Usually, especially for new products	Yes; preservatives and stabilizers	Certain ingredients and additives	Yes; small text and technical terminology	Yes	Yes; wants to know what triggered the warning

Participants also identified other important product information:

- **P1:** Calories, sugar content, expiration date, and price.
- **P2:** Price, protein, calories, and expiration date.
- **P3:** Ingredient quality, additives, oils, and general product composition.
- **P4:** Product rating, additives, origin, production and expiration dates, alternatives, and dietary filters.
- **P5:** Nutritional values, calories, macronutrients, shelf life, and degree of processing.

2.4 Key Findings

The discussions revealed several recurring difficulties related to food ingredient information.

First, all five participants reported that unfamiliar or technical ingredient names can be difficult to understand. Common examples included E-number additives, preservatives, stabilizers, and other complex ingredient descriptions. Some participants also mentioned small or unclear label text. These findings suggest that displaying the original ingredient list alone may not always be sufficient for users to interpret product composition.

Second, all five participants described situations in which they were uncertain whether a product contained an ingredient they wanted to avoid. Their concerns included unfamiliar terminology, unclear packaging, and ingredient names that were difficult to recognize. The ingredients mentioned were not limited to allergens and also included sugar, additives, and ingredients avoided for dietary or personal reasons.

Third, all five participants considered automatic identification of potential allergens useful. Several participants explained that this functionality could reduce the time required to examine ingredient lists manually. One participant also mentioned its potential usefulness for family members with food allergies.

Another consistent finding was the importance of explainability. All five participants preferred an application that identifies the specific ingredient responsible for an allergen warning. Their responses suggest that a generic warning would provide less useful information than a result supported by a recognizable ingredient or ingredient phrase.

Participants also expressed interest in additional product information, including nutritional values, expiration dates, product ratings, pricing, and dietary compatibility. These responses indicate broader consumer interest in food information services. However, such features are not the primary focus of the diploma project.

Although the sample is small, the consistency of the responses provides preliminary qualitative evidence that ingredient interpretation and clear allergen explanations are relevant user needs.

2.5 Influence on Project Requirements

The discussions provided supporting evidence for several requirements already defined for the proposed TazaScan allergen analysis module.

Difficulties understanding unfamiliar ingredient names support the need for ingredient text normalization and NLP-assisted analysis, as addressed by FR-02 and FR-03.

Reported uncertainty about ingredient presence supports the need to handle missing or ambiguous ingredient information carefully, as described in FR-05.

The preference expressed by all five participants for identifying the ingredient responsible for a warning directly supports FR-04 and NFR-01, which require ingredient-level evidence and understandable explanations.

The results did not require a major change to the project scope. TazaScan will continue to provide general food product information, while the diploma contribution remains focused on explainable potential allergen detection and its evaluation. Additional suggestions, such as personalized dietary filters and alternative product recommendations, may be considered for future development but are not included in the current diploma scope.

3. Literature Review

3.1 Food Allergen Information and Label Interpretation

Food labels are the primary source of ingredient and allergen information available to consumers of packaged food. However, the presence of allergen information on a package does not necessarily make it easy to understand. Holleman et al. [1] investigated how allergic and non-allergic consumers interpret allergen information on packaged foods. Using 18 food products, the study found that fewer than half of the participants considered the allergen information clear. The wording of precautionary allergen labels also affected how consumers interpreted the potential risk. PubMed

Blom et al. [2] examined 288 packaged food labels and identified considerable variation in the way allergens were highlighted in ingredient lists and precautionary statements. The study showed that the usability of allergen information depends not only on whether the information is present, but also on how clearly it is communicated. Wiley Online Library

A separate issue is precautionary allergen labelling, such as “may contain.” Turner et al. [3] explain that these statements refer to possible unintended allergen presence rather than an allergen deliberately used as an ingredient. Their review also notes that the use of such labels remains inconsistent. For an automated application, this distinction is important because an allergen explicitly present in the ingredient list should not be represented in the same way as a possible trace. ScienceDirect

These findings influence the proposed TazaScan design in two ways. First, the application should not provide only a binary warning. It should show the ingredient or phrase that caused the result. Second, direct ingredient evidence, insufficient information, and precautionary statements should be represented separately rather than combined into a single “safe” or “unsafe” output.

3.2 Food Ingredient Text Processing and NLP

Automatic allergen detection depends on reliable processing of food ingredient text. Ingredient lists contain alternative names, compound ingredients, abbreviations, spelling variations, and domain-specific terminology. As a result, simple exact keyword matching may not cover all relevant cases.

Popovski et al. [4] developed the FoodBase corpus, a manually checked resource containing annotated food entities. The work demonstrates the importance of domain-specific annotated data for food-related Natural Language Processing tasks and provides a foundation for developing and evaluating food entity extraction methods. PubMed

Stojanov et al. [5] used FoodBase to develop FoodNER, a collection of BERT-based Named Entity Recognition models. The models achieved macro F1-scores between 93.30% and 94.31% when distinguishing food from non-food entities. However, performance decreased to approximately 73.39–78.96% for more detailed semantic classification. This difference is relevant to allergen detection because identifying that a phrase represents food is easier than correctly determining its specific semantic role or allergen relationship. ScienceDirect

Hu et al. [6] applied pretrained language models and machine-learning approaches to large food-label datasets for automated food categorization and nutrition quality prediction. Although their task was not allergen identification, their results show that contextual language models can process food-label text at scale and can provide advantages over simpler text representations. ScienceDirect

These studies support the use of NLP for handling variable ingredient descriptions. At the same time, they do not provide a reason to replace deterministic matching completely. For TazaScan, rule-based analysis remains suitable for clear and known allergen terms, while NLP can be applied to descriptions that are more difficult to interpret.

3.3 Automated Allergen Detection Approaches

Several studies address automatic allergen detection more directly.

Roither et al. [7] investigated automatic allergen and style classification from recipe ingredients. Their work demonstrates that allergen categories can be inferred from ingredient information. However, recipes are usually more structured than commercial ingredient labels and do not fully represent problems such as nested ingredients, unusual terminology, and precautionary statements.

Gadelha et al. [8] developed an NLP-based approach for identifying allergens on Brazilian food labels. Their system combined an allergen terminology database, text preprocessing, and Named Entity Recognition for different names associated with four major allergen groups. The reported evaluation achieved an average precision of 96.50%. The study is particularly relevant to TazaScan because it shows that one allergen may appear under several different ingredient names. At the same time, the authors worked with a limited allergen set and language-specific terminology, which reduces direct generalizability to other markets. jhi.sbis.org.br

Kına [9] studied real-time food allergen detection using OCR together with machine-learning techniques. The work evaluated both clean textual input and a real-time pipeline based on food-label images. The study reports that performance under practical OCR conditions was lower than results obtained using cleaner offline data. This is important for TazaScan because its photo-based product workflow can also introduce OCR errors before allergen analysis begins. OpenAlex

Badli et al. [10] developed a Flutter-based mobile application for packaged food allergen detection using OCR, NLP processing, and a curated allergen-name database. Their evaluation used 40 food-label images and reported a Character Error Rate of 4.76% and Word Error Rate of 13.72%. The work is technically close to TazaScan because it demonstrates a complete mobile scanning pipeline, but it also shows that allergen detection remains dependent on OCR quality and the coverage of the underlying allergen terminology database. DOI

Susanto et al. [11] investigated allergen detection from food-label text as a Named Entity Recognition (NER) task using transformer-based models, including IndoBERT, multilingual BERT (mBERT), and XLM-RoBERTa. The best-performing model, mBERT, achieved an entity-level F1-score of 0.686. The study demonstrates that contextual language models can recognize allergen expressions in food-label text, but their performance remains imperfect and depends on

the quality of annotated data. These findings support the use of NLP for complex ingredient descriptions while highlighting the importance of comparing it with more predictable rule-based methods.

3.4 Hybrid and Explainable Ingredient Analysis

The reviewed methods reveal a trade-off between deterministic and NLP-based approaches. Rule-based or database-based matching is transparent and predictable, but its performance depends on the completeness of predefined terms and synonyms. NLP models can handle more diverse language, but they can introduce false positive and false negative results and are more difficult to explain.

A relevant methodological example is Food Additive Lens by Feng et al. [12]. Although this work focuses on food additives rather than allergens, the system uses a hybrid identification approach that combines direct database lookup with AI-based extraction. The hybrid identifier achieved an F1-score of 0.757 and was combined with an explanation component designed to make results understandable to users. ScienceDirect

Baryannis et al. [13] approach allergen information from a structured knowledge perspective through the Food Allergen Traceability Ontology. Their work shows that allergens, ingredient declarations, products, and evidence can be represented through explicit relationships instead of treating allergen information as isolated keywords. Sage Journals

These approaches support the architecture selected for TazaScan. Clearly identifiable ingredients and known synonyms can be processed using deterministic rules, while NLP processing can be used for more complex or variable descriptions. The final result should remain linked to the ingredient evidence that triggered the detection, regardless of which component produced it.

3.5 Critical Analysis and Research Gap

The literature shows that automatic allergen analysis is already technically feasible. Previous studies have applied terminology databases, rule-based matching, traditional machine learning, Named Entity Recognition, transformer models, OCR, and mobile applications. Therefore, the contribution of this diploma project is not based on the claim that allergen detection has never been implemented before.

However, several limitations remain relevant to the proposed TazaScan module.

First, some existing approaches depend heavily on manually defined terminology or cover a limited set of allergens. This can cause problems when ingredient names differ from the expected dictionary representation. Gadelha et al. [8], for example, focused on four major allergen groups and required language-specific processing. jhi.sbis.org.br

Second, NLP performance is strongly influenced by the availability of annotated domain data. FoodNER demonstrates high performance for broad food entity detection, while more detailed semantic classification is substantially more difficult. ScienceDirect

Third, real-world application performance can be affected before allergen analysis even begins. OCR-based studies show that errors introduced when extracting text from packaging can reduce the reliability of the final pipeline. DOI

Finally, label-communication research demonstrates that the technical detection itself is only part of the problem. A user also needs to understand why a result was produced and whether it represents a directly declared ingredient or only uncertain or precautionary information. PubMed Central (PMC)

The closest studies also leave several limitations relevant to this project. Gadelha et al. [8] demonstrated strong NLP-based allergen recognition, but their approach focused on four allergen groups and language-specific terminology. Badli et al. [10] developed a Flutter-based mobile system combining OCR and allergen identification, but detection depended on matching extracted text against a curated allergen database. Susanto et al. [11] investigated transformer-based allergen extraction and reported a best entity-level F1-score of 0.686, indicating that contextual NLP still faces recognition errors. These findings motivate investigating whether deterministic rules and NLP-based processing can complement each other, rather than relying on either approach alone.

Based on these limitations, the diploma project investigates a **hybrid allergen detection approach** within the existing TazaScan platform. The proposed method combines deterministic rules for clearly identifiable allergen terms with NLP-based processing for more complex or variable ingredient descriptions. Each detected potential allergen will be linked to the ingredient or ingredient phrase that caused the result. The system will distinguish between direct ingredient evidence, precautionary information such as possible traces, and cases where ingredient information is missing or insufficient. These cases will be represented using an explicit result status rather than being reduced to a binary “safe” or “unsafe” conclusion.

The developed approach will also be evaluated quantitatively. A labeled food-ingredient dataset will be prepared, and the hybrid approach will be compared with a rule-based baseline using the same evaluation conditions. Precision, Recall, and F1-score will be used to measure detection performance. This makes it possible to determine whether the additional NLP component provides a measurable improvement rather than assuming that a more complex method is automatically better.

For the local context of TazaScan, the system design will also take into account TR CU 022/2011 “Food Products in Terms of Their Labelling”, which establishes food-labelling requirements applicable in Kazakhstan. The regulation requires specified components that may cause allergic reactions or are contraindicated for particular diseases to be declared in the ingredient list regardless of their quantity. It also addresses situations where such components were not intentionally used but their possible presence cannot be completely excluded [14]. This distinction is relevant when defining direct allergen evidence and precautionary result statuses in TazaScan. However, the application will remain an informational tool rather than a medical or regulatory compliance system.

3.6 Structured Literature Analysis

Ref.	Year	Problem	Method	Main Findings	Main Limitation	Relevance to TazaScan
Holleman et al. [1]	2021	Consumers have difficulty interpreting allergen labels	Controlled experiments with packaged foods	Allergen information was unclear to many participants	Does not develop an automated detector	Supports understandable result presentation
Blom et al. [2]	2021	Inconsistent allergen communication on labels	Analysis of 288 packaged food labels	Strong variation in allergen presentation and PAL	Specific retail/regulatory context	Supports consistent ingredient-level explanations
Turner et al. [3]	2024	Inconsistent use of precautionary allergen labelling	Review of PAL practices	PAL represents possible unintended presence and should be distinguished from declared ingredients	Review rather than software evaluation	Supports separate result status
Popovski et al. [4]	2019	Lack of food-domain annotated NLP data	Development of FoodBase corpus	Demonstrates value of manually annotated food entities	Not allergen-specific	Supports labeled data preparation
Stojanov et al. [5]	2021	Extraction and semantic classification of food entities	BERT-based NER	High food/non-food F1; lower performance for detailed semantics	Not specifically allergen detection	Supports NLP component and demonstrates semantic difficulty
Hu et al. [6]	2023	Automated processing of large food-label datasets	Pretrained NLP and ML models	Contextual NLP can process food-label text effectively	Tasks concern categories/nutrition rather than allergens	Supports NLP over ingredient text
Roither et al. [7]	2022	Automatic allergen classification	Ingredient-based classification	Allergens can be classified automatically from ingredients	Recipe data differs from packaged food labels	Supports ingredient-based allergen analysis
Gadelha et al. [8]	2024	Different names used for allergens on labels	Terminology database + preprocessing + NER	Average reported precision of 96.50%	Limited allergen set and language-specific processing	Closely supports normalization and NLP
Kına [9]	2025	Allergen detection under real OCR conditions	OCR + ML/DL classification	Real-world OCR conditions	Limited dataset and OCR dependency	Supports testing input quality and OCR effects

				reduce performance compared with cleaner data		
Badli et al. [10]	2026	Real-time mobile allergen detection	Flutter + OCR + NLP + allergen database	Demonstrates functioning mobile label-scanning workflow	40 label images; depends on terminology database	Closest mobile-system comparison
Susanto et al. [11]	2026	Allergen extraction from variable label text	Transformer-based NER	Contextual models can identify allergen entities	Probabilistic output and dependence on annotated data	Supports NLP for difficult descriptions
Feng et al. [12]	2026	Identification and explanation of food ingredients/additives	Hybrid database + AI extraction	Hybrid identifier combines deterministic and AI analysis	Focuses on additives rather than allergens	Methodological support for hybrid design
Baryannis et al. [13]	2026	Structured allergen information and traceability	Food allergen ontology	Models relationships between allergens, declarations and evidence	Focus is traceability	Supports structured allergen/evidence representation

4. Existing System Comparison

A comparison of existing food scanning and ingredient-analysis applications was conducted to understand which functions are already available and which limitations are relevant to TazaScan. Four systems were selected: Yuka, Open Food Facts, Spoonful, and Fig.

The systems were compared using the same criteria: barcode scanning, ingredient information, nutrition information, allergen-related analysis, ingredient-level explanation, image or photo-based data input, personalization, product scoring, and important limitations.

Yuka focuses primarily on general food quality assessment. It provides barcode-based product analysis, nutritional information, a 0–100 food score, and recommendations for better-rated alternatives. Its Premium food preferences can warn users about selected ingredients such as gluten, lactose, soy, and sulfites, but Yuka explicitly states that precautionary traces such as “may contain” are not taken into account. help.yuka.io

Open Food Facts is an open collaborative food database and scanning application. It provides barcode scanning, ingredients, allergens, nutrition facts, Nutri-Score, and other food attributes. The platform can also use personal food preferences to highlight products that match or conflict with selected criteria. However, the information depends heavily on the completeness and accuracy of collaboratively submitted product data. blog.openfoodfacts.org

Spoonful is more focused on dietary restrictions, allergies, and intolerances. Users can select supported diets and allergens and scan products to check whether they fit those restrictions. Spoonful analyzes ingredient lists and allergen statements and can highlight ingredients that cause a warning. It also supports photo-based correction or submission of ingredient information. The application notes that it cannot fully determine manufacturing cross-contamination risks. Spoonful

Fig is built around a personalized dietary profile containing allergies, intolerances, diets, and individual ingredients to avoid. The application supports barcode scanning and compares product ingredients against the user's selected restrictions. Users can inspect an ingredient to understand why it does not match their profile. Fig provides strong personalization, but its current consumer availability is primarily limited to the United States.

4.1 Comparison Table

Criterion	**Yuka**	**Open Food Facts**	**Spoonful**	**Fig**
Barcode scanning	Yes	Yes	Yes	Yes
Ingredient information	Yes	Yes	Yes	Yes
Nutrition information	Yes	Yes	Yes	Available as product-label information, but not the main focus
Allergen-related analysis	Limited / selected alerts	Allergen data and alerts	Yes, regulated allergens and diet restrictions	Yes, allergies and ingredient restrictions
Ingredient-level explanation	Limited	Limited	Yes, flagged ingredients and notes	Yes, user can inspect why an ingredient does not match
Photo / image-based data input	Product photos can be submitted	Product label photos can be contributed	Yes, label photos can update ingredient information	Not a main documented workflow

Personalization	Dietary preferences in Premium	Custom food preferences	Diet/allergy profile	Strong personalized dietary profile
Product scoring	0–100 Yuka score	Nutri-Score / NOVA	Diet-specific color/result system	Compatibility/color-coded result rather than general food score
Important limitation	Does not account for “may contain” traces	Depends on completeness of community data	Cannot fully assess manufacturing cross-contamination	Mainly available in the U.S.

4.2 Findings and Relevance to TazaScan

The comparison shows that barcode scanning, ingredient display, dietary filtering, and allergen-related warnings already exist in several applications. Therefore, these functions alone are not sufficient to define the technical contribution of TazaScan.

For the diploma project, the focus is placed on the implementation and evaluation of an explainable allergen detection module within the existing TazaScan platform. The proposed module will combine deterministic ingredient rules with NLP-based processing for more complex ingredient descriptions and will return the ingredient evidence responsible for each detected potential allergen.

Unlike systems whose main purpose is personalized dietary matching, the diploma version of TazaScan does not build a personal diet profile or generate nutrition recommendations. Its main research and engineering focus is the detection of potential allergens from ingredient information and the quantitative evaluation of this detection method.

5. Technology Comparison

The main technical decisions were reviewed against reasonable alternatives. Since TazaScan already has a working codebase, the comparison considers not only general advantages of each technology but also compatibility with the existing system, implementation effort, maintainability, and suitability for the planned allergen analysis module.

5.1 Mobile Application Framework

Option	Advantages	Limitations	Decision
Flutter	One codebase for multiple platforms, mature UI toolkit, already used in TazaScan	Larger application size than some native solutions	**Selected**
React Native	Cross-platform development, large JavaScript ecosystem	Would require rewriting the existing Flutter application	Not selected
Native Android / Kotlin	Strong Android integration and native performance	Separate development would be required for other platforms and the current Flutter application would need to be replaced	Not selected

Flutter was retained because the existing TazaScan mobile application is already implemented with it. Replacing the framework would add significant development work without providing a clear benefit for the diploma objective.

5.2 Backend Platform

Option	Advantages	Limitations	Decision
Node.js + Express + TypeScript	Suitable for REST APIs, existing TazaScan backend, shared TypeScript ecosystem with the admin application	CPU-intensive processing may require additional handling	**Selected**
Spring Boot	Mature architecture and strong enterprise tooling	Higher migration effort and would require rewriting the current backend	Not selected
FastAPI	Convenient for Python-based ML/NLP integration and rapid API development	Would introduce another backend stack and duplicate existing API functionality	Not selected

5.3 Database Technology

Option	Advantages	Limitations	Decision

PostgreSQL	Strong relational model, transactions, constraints, good Prisma support	Schema changes require migrations	**Selected**
MySQL	Mature relational database and widely supported	No significant advantage over the existing PostgreSQL setup for this project	Not selected
MongoDB	Flexible document structure	Less suitable for the existing relational product, submission, certification, and user data model	Not selected

PostgreSQL was retained because TazaScan already uses a relational data model containing products, product submissions, users, product families, and Halal certification records. Prisma is already used as the data-access layer, so changing the database would add unnecessary migration work.

5.4 Allergen Detection Approach

Option	Advantages	Limitations	Decision
Rule-based only	Predictable, easy to explain, effective for known ingredient names and synonyms	May struggle with complex or ambiguous ingredient descriptions	Baseline
NLP-based only	Can potentially handle more complex language patterns	Less predictable and more difficult to explain; depends on data/model quality	Not selected as standalone approach
Hybrid approach	Combines explainable rules for clear cases with NLP processing for more complex cases	More implementation and evaluation work	**Proposed approach**

Node.js with Express and TypeScript was retained because the existing TazaScan backend already supports product management, authentication, scoring, OCR processing, Halal screening, and other application services. The planned allergen detection module can be integrated into this architecture through the existing REST API without rebuilding the backend. This approach

reduces migration effort, preserves the current functionality, and allows development to focus on the allergen analysis module.

A hybrid approach is proposed for the diploma project. Rule-based detection will provide an explainable baseline for clearly identifiable ingredients, while NLP-based processing will be investigated for more complex ingredient descriptions. The final evaluation will determine whether the hybrid approach provides an improvement over the rule-based baseline.

5.5 Technology Selection Summary

The selected technologies preserve the existing TazaScan architecture and allow the diploma work to focus on the new allergen analysis functionality instead of rebuilding already implemented components. Flutter, Node.js/Express, PostgreSQL, and Prisma therefore remain part of the existing platform, while the hybrid allergen detection approach represents the main new technical decision introduced by the diploma project.

6. Requirements Specification

The requirements below describe the functionality that must be supported by the diploma version of TazaScan. The main focus is the integration of potential allergen detection into the existing product analysis workflow.

6.1 Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	The system shall allow the user to open a product through the existing barcode, text search, or package photo flow.	Must	A product found through any supported search method can be opened in the product details screen.
FR-02	The system shall use the available ingredient information of a product as input for allergen analysis.	Must	When ingredient data is available, it is passed to the allergen analysis module.

FR-03	The system shall identify potential allergens using a hybrid approach combining rule-based analysis and NLP-based processing.	Must	The module returns potential allergens for ingredient lists used in the evaluation dataset.
FR-04	The system shall provide evidence for each detected allergen.	Must	Each detected allergen is linked to the ingredient or ingredient phrase that caused the detection.
FR-05	The system shall distinguish between a detected allergen and a case where there is not enough ingredient information for reliable analysis.	Must	Products with missing or insufficient ingredient data do not receive an unsupported allergen result.
FR-06	The allergen analysis result shall be displayed as part of the existing TazaScan product details flow.	Must	The user can view the allergen result without leaving the product analysis flow.
FR-07	The system shall preserve the existing product information functionality, including nutrition information, ingredients, TazaScore, Nutri-Score, and Halal Check.	Must	Existing product information remains available after integration of the allergen module.
FR-08	The developed allergen detection method shall be testable using a labeled evaluation dataset.	Must	The module can be executed on the prepared dataset and its predictions can be compared with known labels.
FR-09	The system should provide a confidence or result-status indicator where the available ingredient information is incomplete or ambiguous.	Should	The UI distinguishes normal detections from cases that require caution or additional review.

6.2 Non-Functional Requirements

NFR-01 — Explainability
The system shall present allergen detection results in a way that allows the user to understand which ingredient caused each result.

NFR-02 — Reliability
The system shall avoid presenting a definite allergen result when the available ingredient information is missing or insufficient.

NFR-03 — Maintainability
Allergen detection rules, allergen categories, and NLP-related components should be organized so that they can be updated without major changes to the rest of the TazaScan system.

NFR-04 — Compatibility
The allergen module shall be integrated with the existing TazaScan backend and Flutter mobile application without removing the current product analysis functionality.

NFR-05 — Testability
The allergen detection logic shall support automated testing and quantitative evaluation using Precision, Recall, and F1-score.

NFR-06 — Safety of Communication
The application shall present allergen results as informational detection of potential allergens and shall not present them as medical diagnosis or a guarantee that a product is safe.

7. Architecture and Diagrams

TazaScan follows a client-server architecture. The system consists of a Flutter mobile application for consumers, a React-based web administration panel, a Node.js/Express backend, and a PostgreSQL database accessed through Prisma.

The mobile application communicates with the backend through REST API endpoints. It is responsible for barcode scanning, text and photo-based product search, displaying product information, ingredients, nutrition data, TazaScore, Nutri-Score, Halal Check, history, favorites, and product submission functionality.

The administration panel uses the same backend and provides functionality for product management, product image management, Halal certification management, and moderation of user product submissions.

The backend contains the main application logic, including product services, scoring engines, Halal screening, OCR processing, image processing, data-quality checks, authentication, and access to persistent data through Prisma and PostgreSQL.

The planned allergen detection module will be added as a separate backend component and integrated into the existing product analysis workflow. It will receive ingredient information, preprocess and normalize the ingredient text, apply rule-based and NLP-based analysis, combine the results, and return potential allergens together with supporting ingredient evidence and result status.

Open Food Facts is used only through controlled import scripts to populate or update product data. It is not queried directly during the normal consumer product lookup flow.

7.1 System Context Diagram

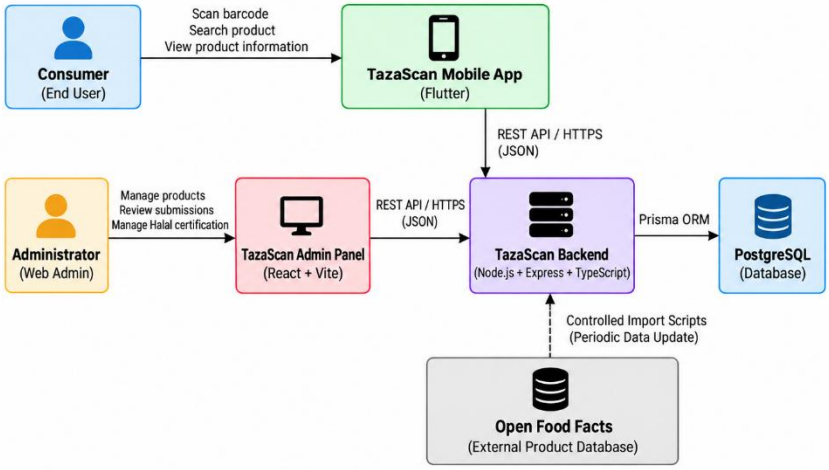


Figure 1. TazaScan System Context Diagram

7.2 Use Case Diagram

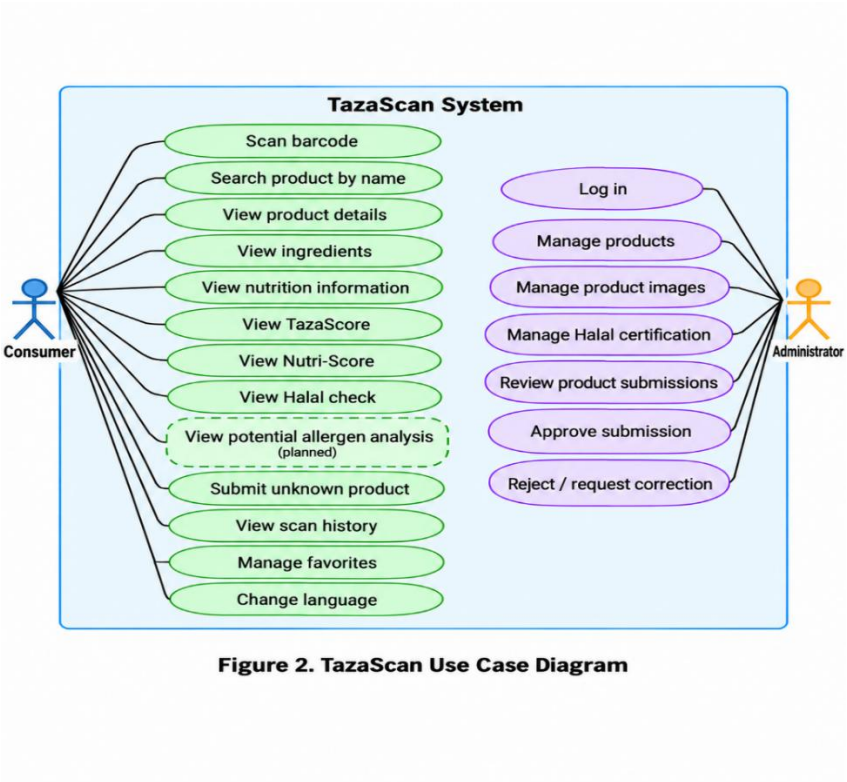


Figure 2. TazaScan Use Case Diagram

7.3 Component Diagram

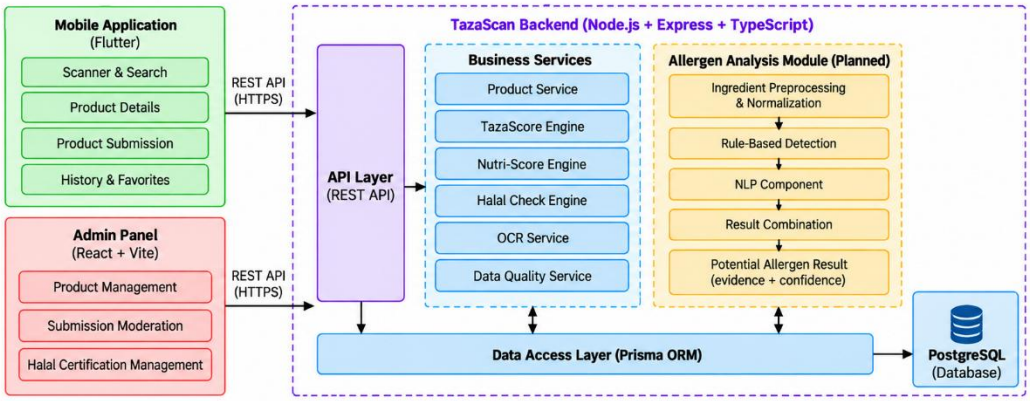


Figure 3. TazaScan Component Diagram

The proposed allergen analysis module is separated from the existing TazaScore, Nutri-Score, Halal Check, and OCR components. This allows the new functionality to be developed and evaluated independently while using the existing TazaScan product and ingredient data.

8. UX / Prototype

TazaScan already has a functional mobile prototype and a web administration interface. Therefore, this section is based primarily on the implemented user interface rather than conceptual mockups.

The main consumer flow starts with product identification. A user can scan a barcode, enter a barcode manually, search for a product by name, or use a package photo. After a product is found, the application opens the product details screen, where the user can view nutrition information, ingredients, TazaScore, Nutri-Score, Halal Check, and information about the reliability of the available product data.

If a product is not available in the database, the user can submit it for moderation together with product information and package images. The application also provides history and favorites for previously viewed products.

The planned allergen analysis functionality will be integrated into the existing product details flow. The user will be able to view detected potential allergens together with the ingredient or ingredient phrase that caused each result. The new functionality will extend the existing product analysis interface rather than introducing a separate application flow.

8.1 Product Search and Identification

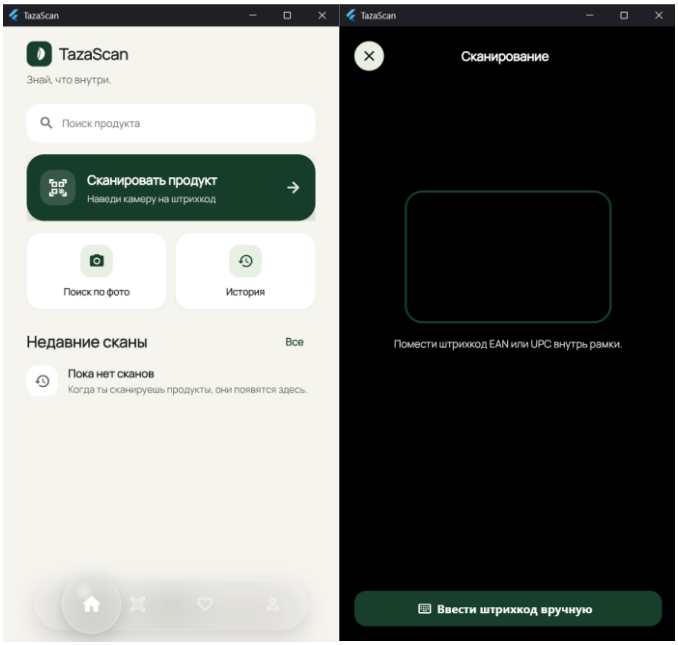


Figure 4. Product identification using barcode scanning

8.2 Product Details

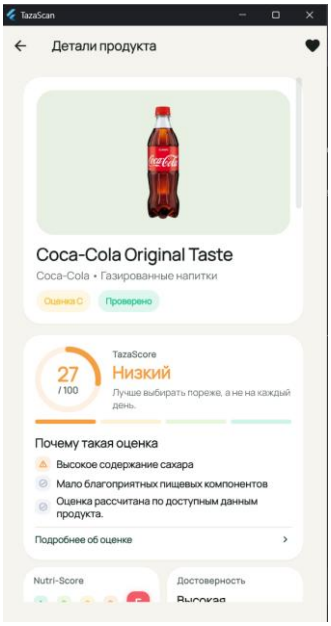


Figure 5. TazaScan product details screen

8.3 TazaScore Details

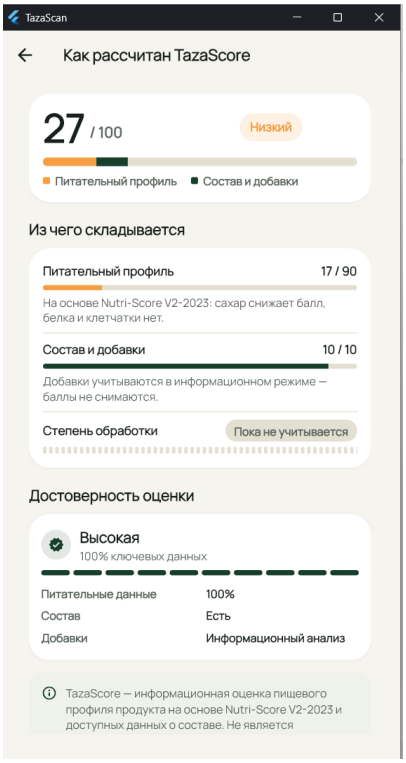


Figure 6. Detailed TazaScore explanation

8.4 Ingredient Information

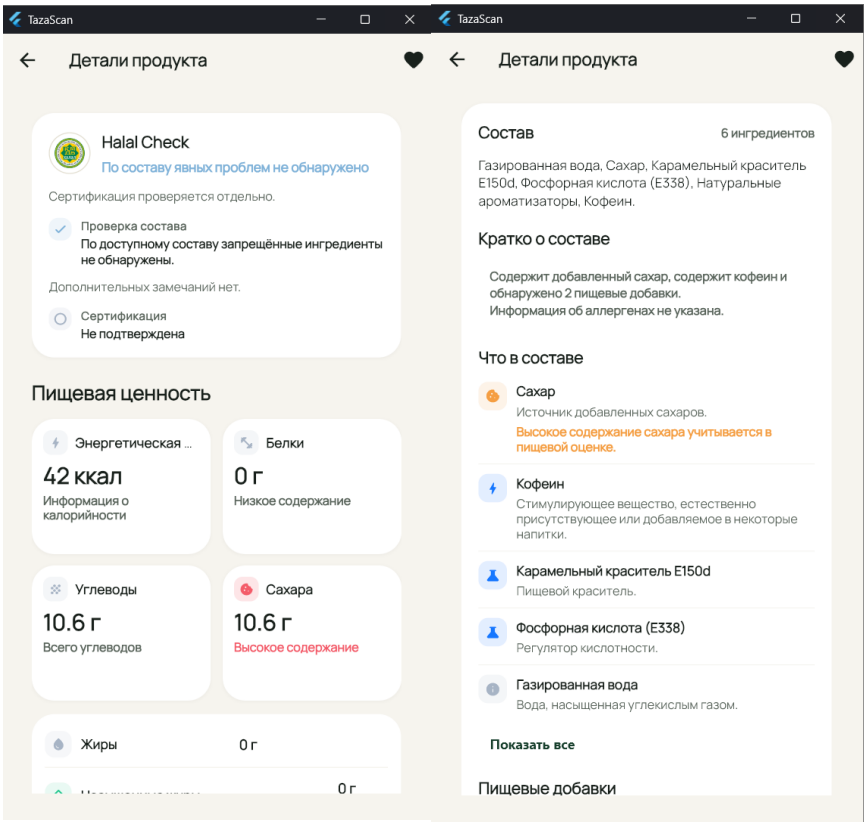


Figure 7. Ingredient information and Halal Check

8.5 Product Submission

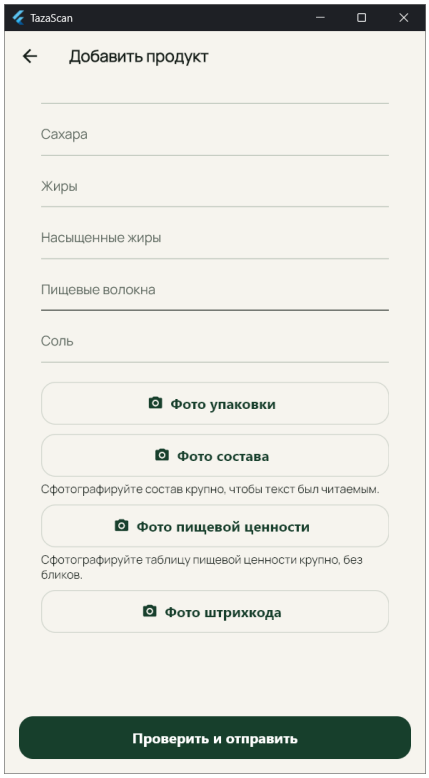


Figure 8. Unknown product submission flow

Planned allergen analysis interface

A dedicated allergen result section will be added to the product details screen during the diploma implementation stage. The interface is planned to display the potential allergen, supporting ingredient evidence, and result status.

9. Verification and Evaluation Plan

The verification and evaluation will focus on both the technical integration of the allergen module and the quality of its detection results.

A labeled evaluation dataset will be prepared using product ingredient lists with known allergen information. The same allergen categories and preprocessing rules will be used for all evaluated approaches.

The main baseline will be a rule-based allergen detector that identifies allergens using predefined ingredient names, synonyms, and matching rules. The proposed hybrid approach will be evaluated under the same conditions and compared with this baseline.

Detection quality will be measured using **Precision, Recall, and F1-score**. These metrics will be used to determine how often the system correctly identifies allergens, how many relevant allergens it successfully detects, and the overall balance between missed and incorrect detections.

In addition to detection quality, functional testing will verify that the allergen module is correctly integrated into TazaScan and that the result shown to the user contains supporting ingredient evidence.

Evaluation Area	Verification Method	Expected Result
Allergen detection	Test on labeled ingredient dataset	Potential allergens are identified correctly
Precision	Compare predictions with known labels	Measure false-positive behavior
Recall	Compare predictions with known labels	Measure missed allergens
F1-score	Calculate from Precision and Recall	Compare overall performance
Baseline comparison	Rule-based vs Hybrid	Determine whether the hybrid approach improves detection
Evidence	Check detected allergen against triggering ingredient	Each detection includes supporting ingredient evidence
Missing ingredient data	Test products with incomplete/missing composition	System does not present an unsupported definite result
Backend integration	API/integration tests	Allergen result is returned correctly
Mobile integration	UI/functional tests	Result is shown in the existing product details flow
Regression testing	Run existing TazaScan tests	Existing functions continue to work

Measurement Conditions

All compared approaches will be evaluated using the same labeled test data, allergen categories, and evaluation procedure. Products with unavailable ingredient information will be treated separately from products for which no allergen is detected. If the NLP component requires model training or adaptation, the final test data will not be used during model development.

10. Project Management and Risks

TazaScan is an individual diploma project. The student is responsible for the mobile application, backend, database, integration of the allergen detection module, testing, evaluation, and project documentation.

GitHub is used as the main version control system. Development progress, source code, documentation, diagrams, and tests will be maintained in the same repository throughout the

diploma project. YouTrack will be used to organize development tasks, priorities, deadlines, and acceptance criteria.

The project will be developed incrementally. The first stage focuses on project analysis, requirements, architecture, literature review, comparison of existing systems and technologies, and preparation of the evaluation plan. The next stages will cover the design and implementation of the allergen analysis module, its integration into TazaScan, preparation of the labeled evaluation dataset, experimental evaluation, testing, and final documentation.

Risk / Dependency	**Possible Impact**	**Mitigation**
Limited labeled allergen data	Difficult to evaluate the detector reliably	Prepare and verify a dedicated labeled evaluation dataset
Complex or ambiguous ingredient names	Potential allergens may be missed or incorrectly detected	Combine rule-based detection with NLP-based processing
Missing ingredient information	Reliable allergen analysis may not be possible	Return an insufficient-data/result-status message instead of a definite result
NLP component does not improve the baseline	Hybrid approach may not outperform rule-based detection	Evaluate both approaches using the same dataset and report the result honestly
Integration affects existing TazaScan functionality	Existing product analysis features may break	Use automated tests and regression testing during integration
Limited project time	Some secondary functionality may remain unfinished	Keep allergen detection and evaluation as the main diploma priority
Dependence on product data quality	Poor input data can reduce analysis quality	Track data completeness and clearly communicate limitations

Main Milestones

Milestone 1: Project analysis and pre-defense materials — proposal, requirements, literature review, comparisons, architecture, prototype, and evaluation plan.
Milestone 2: Allergen detection design and dataset preparation.
Milestone 3: Rule-based baseline and NLP/hybrid implementation.
Milestone 4: Integration with TazaScan mobile application and backend.
Milestone 5: Experimental evaluation, testing, and analysis of results.
Milestone 6: Final documentation, system refinement, and preparation for defense.

11. Supervisor Feedback

During the discussion of the diploma topic, the supervisor noted that the proposed project had similarities with an already approved 2026 topic related to AI-based food analysis and personalized nutrition recommendations.

The supervisor asked whether personalized nutrition recommendations would also be included in TazaScan. Based on this discussion, the project scope was clarified: personalized nutrition recommendations and meal planning are not part of the diploma project. The diploma work focuses specifically on potential allergen detection based on food ingredient analysis.

The supervisor also emphasized the need to make the distinctive feature and novelty of the project clear. TazaScan remains the existing application platform, while the allergen analysis module is defined as the main new contribution of the diploma project.

Supervisor Feedback	**Action Taken**
The topic may overlap with an approved project related to personalized nutrition recommendations.	Personalized nutrition recommendations were explicitly excluded from the project scope.
The difference between TazaScan and similar projects should be clear.	The diploma focus was narrowed to potential allergen detection based on ingredient analysis.
The project should have a clear distinctive feature and novelty.	The allergen analysis module was defined as the main new diploma contribution within the existing TazaScan platform.

Project Repository and Prototype Resources

Submitted GitHub Version:
https://github.com/AidosAidaruly/TazaScan/tree/prob-period-1

Existing Flutter Mobile Prototype (Source Code):
https://github.com/AidosAidaruly/TazaScan/tree/prob-period-1/apps/mobile

Architecture Diagrams — Editable Mermaid Sources:

System Context Diagram:
https://github.com/AidosAidaruly/TazaScan/blob/prob-period-1/docs/diploma/pre-defense-1/diagrams/system-context.mmd

Use Case Diagram:
https://github.com/AidosAidaruly/TazaScan/blob/prob-period-1/docs/diploma/pre-defense-1/diagrams/use-case.mmd

Component Diagram:
https://github.com/AidosAidaruly/TazaScan/blob/prob-period-1/docs/diploma/pre-defense-1/diagrams/component-diagram.mmd

Project Management Board:
https://tazascan.youtrack.cloud/agiles/204-1/current

The existing TazaScan Flutter application serves as the current functional MVP. The hybrid allergen detection module is planned for the next development stage and is not yet implemented.

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